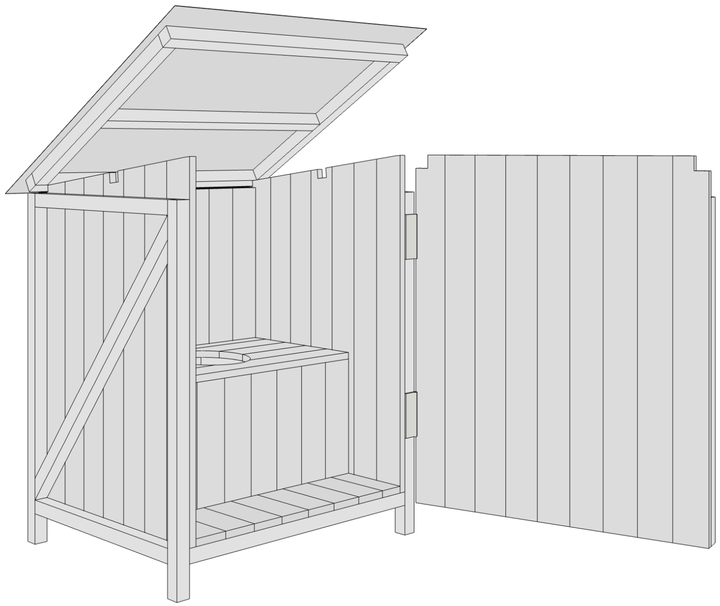
WORKING DRAWING

CAN AI BUILD A TOILET YET?

An experiment in re-drawing a technical drawing as an editable parametric CAD model with Claude and Codex.

PROJECT	DASS
ISSUE	2026-07-30
ARCHITECT	@HANNES.SODERQUIST
CODE	@FEELEPXYZ
SOURCE	GITHUB
UNITS	MILLIMETRES
KERF	2.8 MM PER CUT
SHEETS	A-200 TO A-400



The finished building.

STRUCTURAL TIMBER

MATERIAL

Frame timber, 45 × 45 × 4 200 mm. Quantity: 9 lengths.

The 9 beam stock lengths below are named B01 to B09. Each cut diagram shows the marked beam pieces cut from that length.

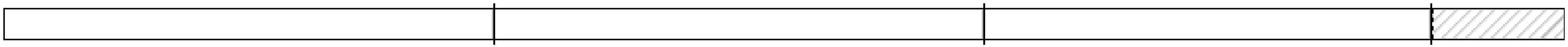
Cut every piece at one stop setting before you change the stop. A batch that reaches the end of a stock length continues on the next length.

These sheets are exact only for this material size. One 2.8 mm kerf is used for every piece.

If your material ends are rough, shave a small amount off from each piece first.

- [ ] B01

3 PIECES



BWD1
1 315.9 mm

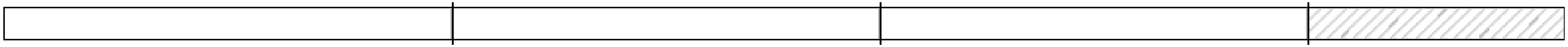
DBD1
1 315.9 mm

LSD1
1 203 mm

356.8 mm

- [ ] B02

3 PIECES



RSD1
1 203 mm

LSV1
1 150 mm

LSV2
1 150 mm

688.6 mm

☐ B03

3 PIECES

RSV1
1 150 mm

RSV2
1 150 mm

DBV1
1 050 mm

841.6 mm

☐ B04

4 PIECES

DBV2
1 050 mm

RBH1
990 mm

RBH2
990 mm

BWH1
900 mm

258.8 mm

☐ B05

4 PIECES

BWH2
900 mm

DBH1
900 mm

DBH2
900 mm

FBH1
900 mm

588.8 mm

☐ B06

4 PIECES

RBC1
900 mm

FBB1
854 mm

SBB1
854 mm

SBB2
854 mm

726.8 mm

☐ B07

5 PIECES

SBF1
854 mm

SBH1
852 mm

SBH2
852 mm

RBS1
812.7 mm

RBS2
812.7 mm

2.7 mm

☐ B08

5 PIECES

LSH1	LSH2	RSH1	RSH2	FBS1	
725 mm	725 mm	725 mm	725 mm	657 mm	629 mm

☐ B09

5 PIECES

FBS2	SBS1	SBS2	SBS3	SBS4	
657 mm	387 mm	387 mm	387 mm	387 mm	1 981 mm

RÅSPONT (MATCHBOARD/V-GROOVE CLADDING)

MATERIAL

Råspont (matchboard/V-groove cladding), 120 × 23 × 4 500 mm.
Quantity: 12 lengths.

The 12 cladding stock lengths below are named P01 to P12. Each cut diagram shows the marked boards cut from that length.

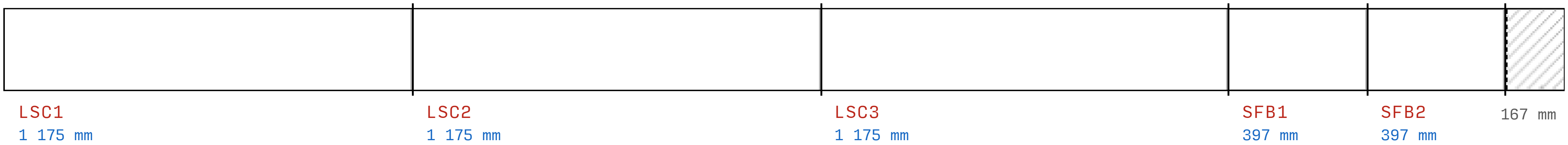
The first 8 stock lengths release all twenty-three 1 175 blanks. Label the P01 to P03 remainders. Keep them for the final 397 pass. Fourteen side boards come next, then nine door boards, at one stop setting.

These sheets are exact only for this material size. One 2.8 mm kerf is used for every piece.

If your material ends are rough, shave a small amount off from each piece first.

☐ P01

5 PIECES



☐ P02

5 PIECES

LSC4	LSC5	LSC6	SFB3	SFB4	
1 175 mm	1 175 mm	1 175 mm	397 mm	397 mm	167 mm

☐ P03

5 PIECES

LSC7	RSC1	RSC2	SFB5	SFB6	
1 175 mm	1 175 mm	1 175 mm	397 mm	397 mm	167 mm

☐ P04

3 PIECES

RSC3	RSC4	RSC5	
1 175 mm	1 175 mm	1 175 mm	966.6 mm

☐ P05

3 PIECES

RSC6	RSC7	DCB1	
1 175 mm	1 175 mm	1 175 mm	966.6 mm

☐ P06

3 PIECES

DCB2 — 1 175 mm | DCB3 — 1 175 mm | DCB4 — 1 175 mm | 966.6 mm

☐ P07

3 PIECES

DCB5 — 1 175 mm | DCB6 — 1 175 mm | DCB7 — 1 175 mm | 966.6 mm

☐ P08

4 PIECES

DCB8 — 1 175 mm | DCB9 — 1 175 mm | BWC1 — 1 050 mm | BWC2 — 1 050 mm | 38.8 mm

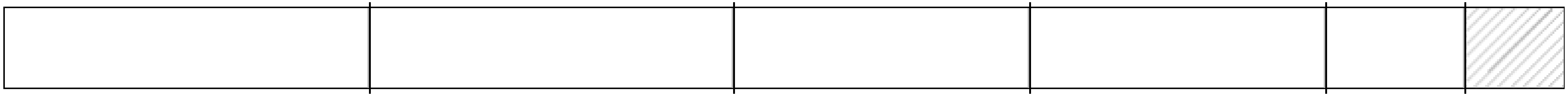
☐ P09

4 PIECES

BWC3 — 1 050 mm | BWC4 — 1 050 mm | BWC5 — 1 050 mm | BWC6 — 1 050 mm | 288.8 mm

☐ P10

5 PIECES



BWC7 1 050 mm | BWC8 1 050 mm | STB1 852 mm | STB2 852 mm | SFB7 397 mm | 285 mm

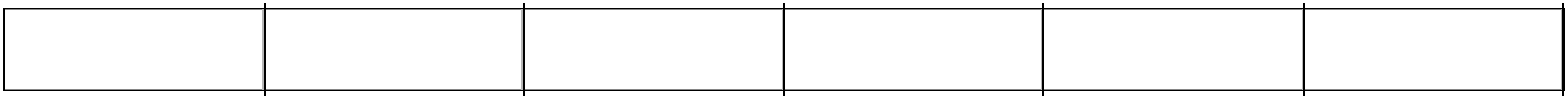
☐ P11

6 PIECES

STB3 852 mm | STB4 852 mm | STB5 852 mm | FCB1 747 mm | FCB2 747 mm | SFB8 397 mm | 36.2 mm

☐ P12

6 PIECES



FCB3 747 mm | FCB4 747 mm | FCB5 747 mm | FCB6 747 mm | FCB7 747 mm | FCB8 747 mm | 1.2 mm

Gaps between boards change the joined width. Fix each board to its unit frame before you mark any final edge. Use the frame edge to set the circular-saw guide. Cut only after the boards are fixed. Sheet A-400 shows each estimated overhang and cut line.

UNIT DRAWINGS AND ASSEMBLY

Each drawing is an orthographic projection off the model. Use the Frame and Cladding controls to inspect the fastening layers and the fitted fields.

Black frame paths are measured 120 mm screws: they start on the source beam and finish in the centre of the receiving section. Diagonal paths follow the diagonal beam from its mitred end.

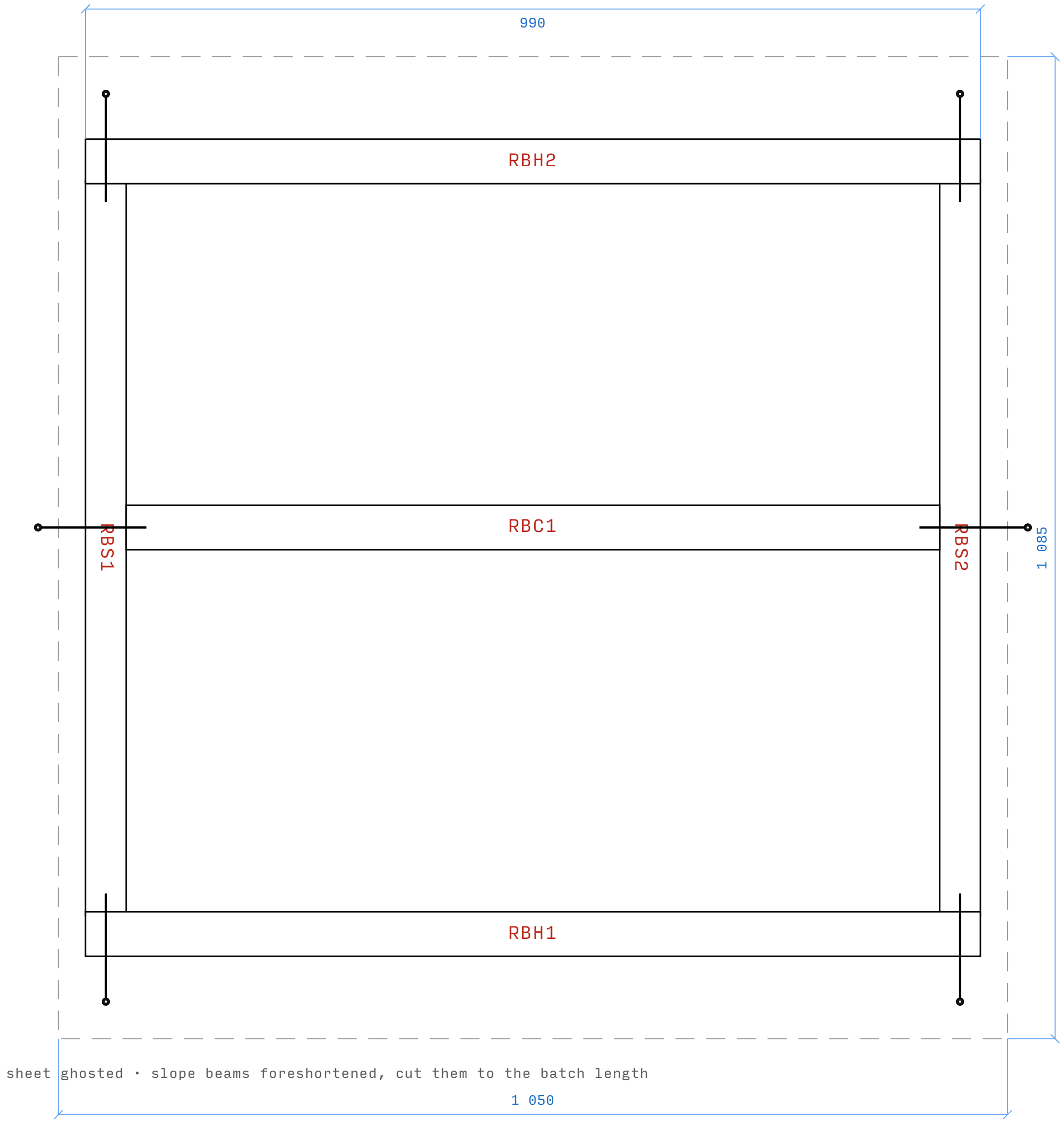
Each unit is drawn on its own sheet, with its assembly steps and its codes beside it.

ASSEMBLY HARDWARE

Use these fasteners for the timber assembly.

USE	FASTENER
Frame beams and braces	6 × 120 mm sunk wood screws
Beam-to-beam support joints, for example floor supports	6 × 90 mm sunk wood screws
Råspont (matchboard/V-groove cladding) to beams	2.8 × 60 mm nails or 6 × 60 mm sunk wood screws

Stack A ROOF UNIT



ROOF UNIT · A-401 · PLAN

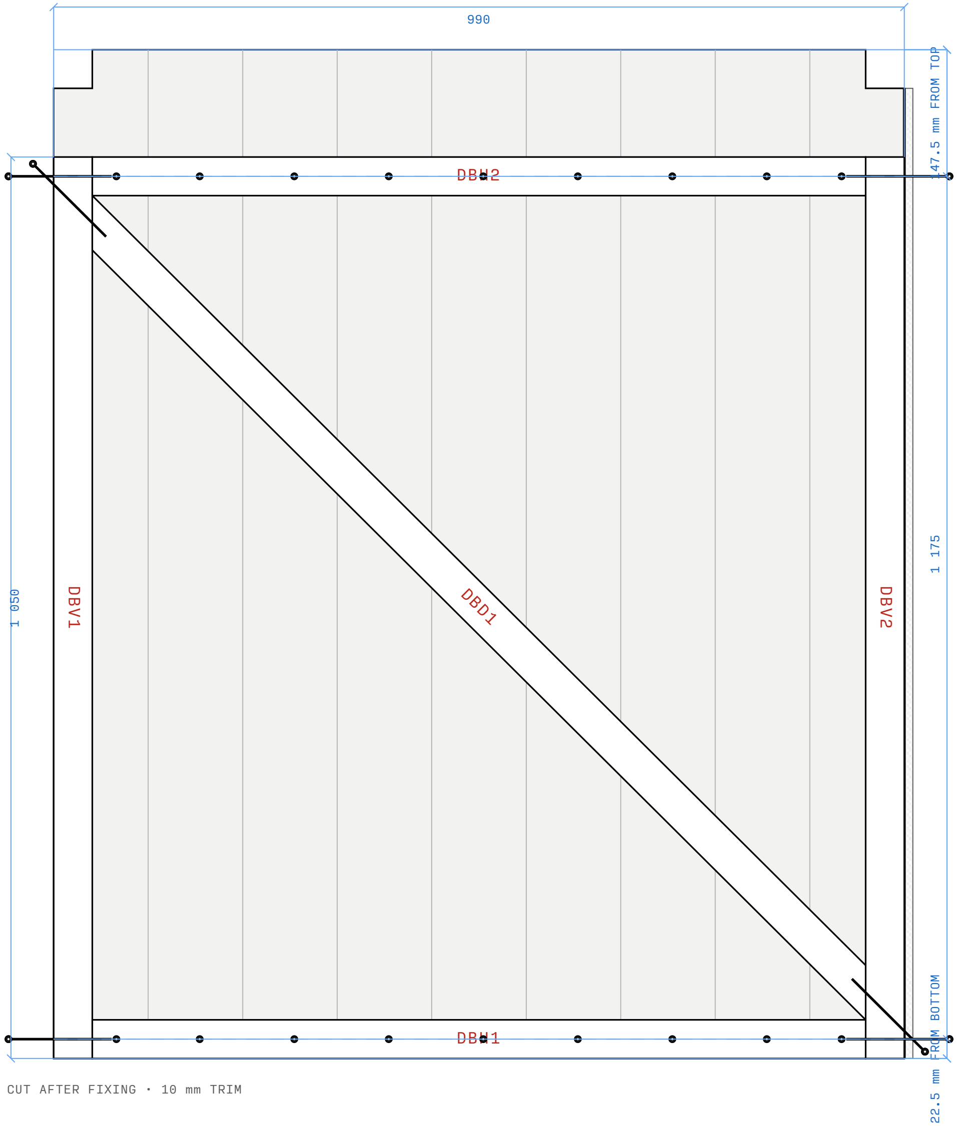
front edge at the bottom · built flat, then hung

ASSEMBLY

1. Lay RBH1 and RBH2 around RBS1 and RBS2. Center RBC1. Match the diagonals, then place the frame screw marks.
2. Center the metal sheet with 30 at each side and 93.1 at the front and rear.
3. Fit the moving hinge leaf. After the shell is square, fit the fixed leaf and hang the roof.

RBC1 RBH1 RBH2 RBS1 RBS2

☐ UNIT COMPLETE



DOOR UNIT · A-402 · FRONT ELEVATION

seen from outside the door · hatched edge is the estimated on-frame trim

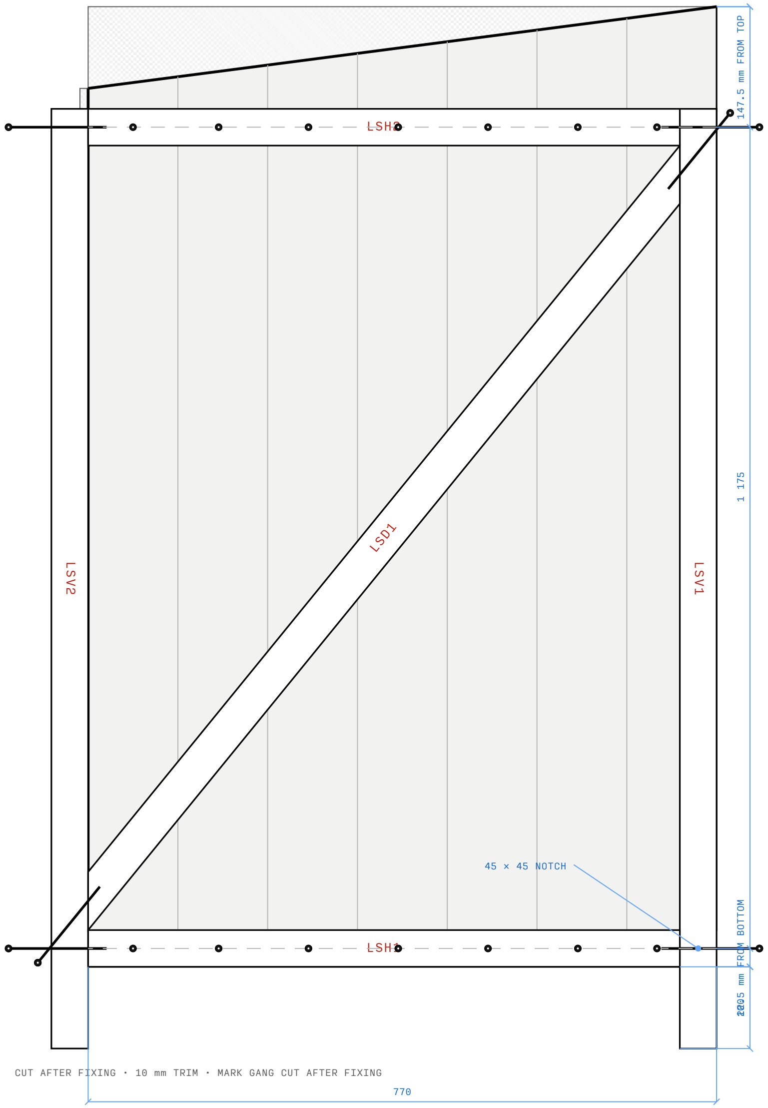
ASSEMBLY

1. Build the DBV and DBH frame. Fit DBD1 and match the diagonals.
2. Fix DCB1 to DCB9 on the inside face. Align DCB1 with the left frame edge.
3. Use the right frame edge to set the circular-saw guide. Cut the estimated 10 overhang after the boards are fixed.
4. Cut the two top reliefs shown. Fit the moving hinge leaves, then hang the door with a 10 gap after the shell is square.

DBD1 DBH1 DBH2 DBV1 DBV2 DCB1 DCB2
DCB3 DCB4 DCB5 DCB6 DCB7 DCB8 DCB9

☐ UNIT COMPLETE

Stack C LEFT SIDE



LEFT SIDE UNIT · A-403 · LEFT ELEVATION

front edge to the right · 100 fall over 770 · 45 × 45 bottom-front notch

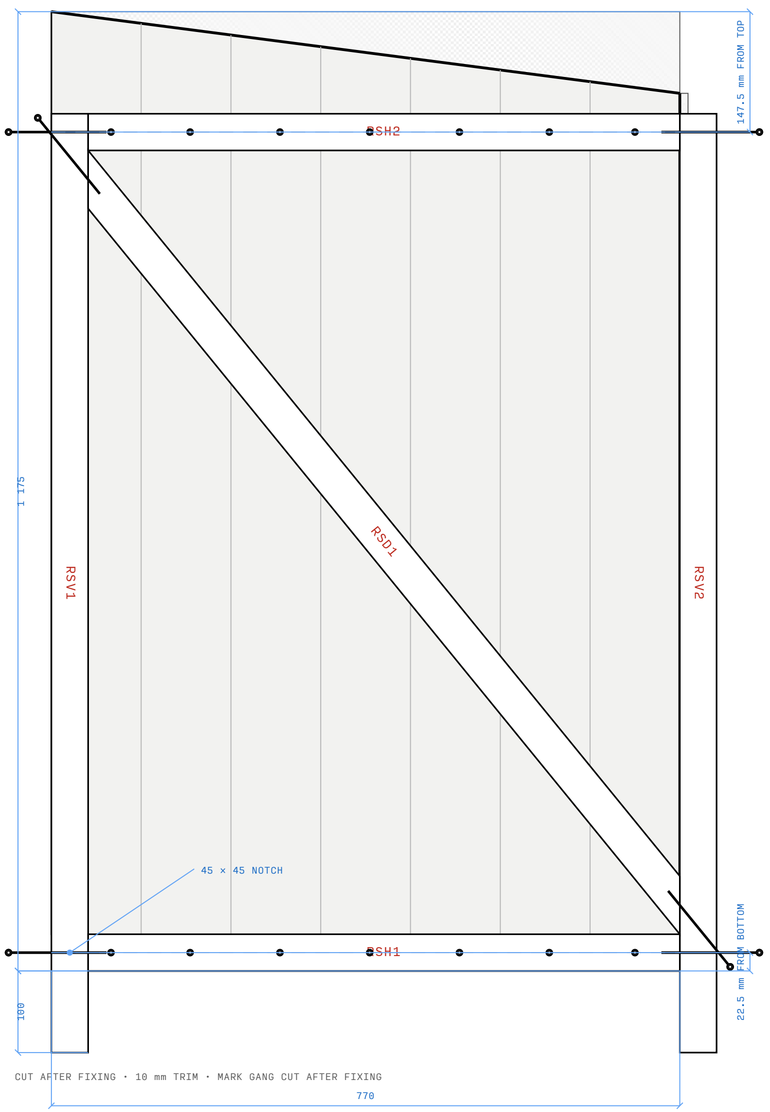
ASSEMBLY

1. Build the frame. Match its diagonals before you fix LSC1 to LSC7.
2. Fix the square LSC boards from front to rear. Align their lower and front edges.
3. Mark 1 175 at the front and 1 075 at the rear. Set the circular-saw guide and make one gang cut after the boards are fixed.
4. Use the rear frame edge to set the circular-saw guide. Cut the estimated 10 overhang, then cut the 45 × 45 bottom-front notch.
5. Do not pre-cut the roof reliefs. Hang and close the roof, then cut only to the scribe.

LSC1 LSC2 LSC3 LSC4 LSC5 LSC6 LSC7
LSD1 LSH1 LSH2 LSV1 LSV2

☐ UNIT COMPLETE

Stack D RIGHT SIDE



RIGHT SIDE UNIT · A-404 · RIGHT ELEVATION

front edge to the left · 100 fall over 770 · 45 × 45 bottom-front notch

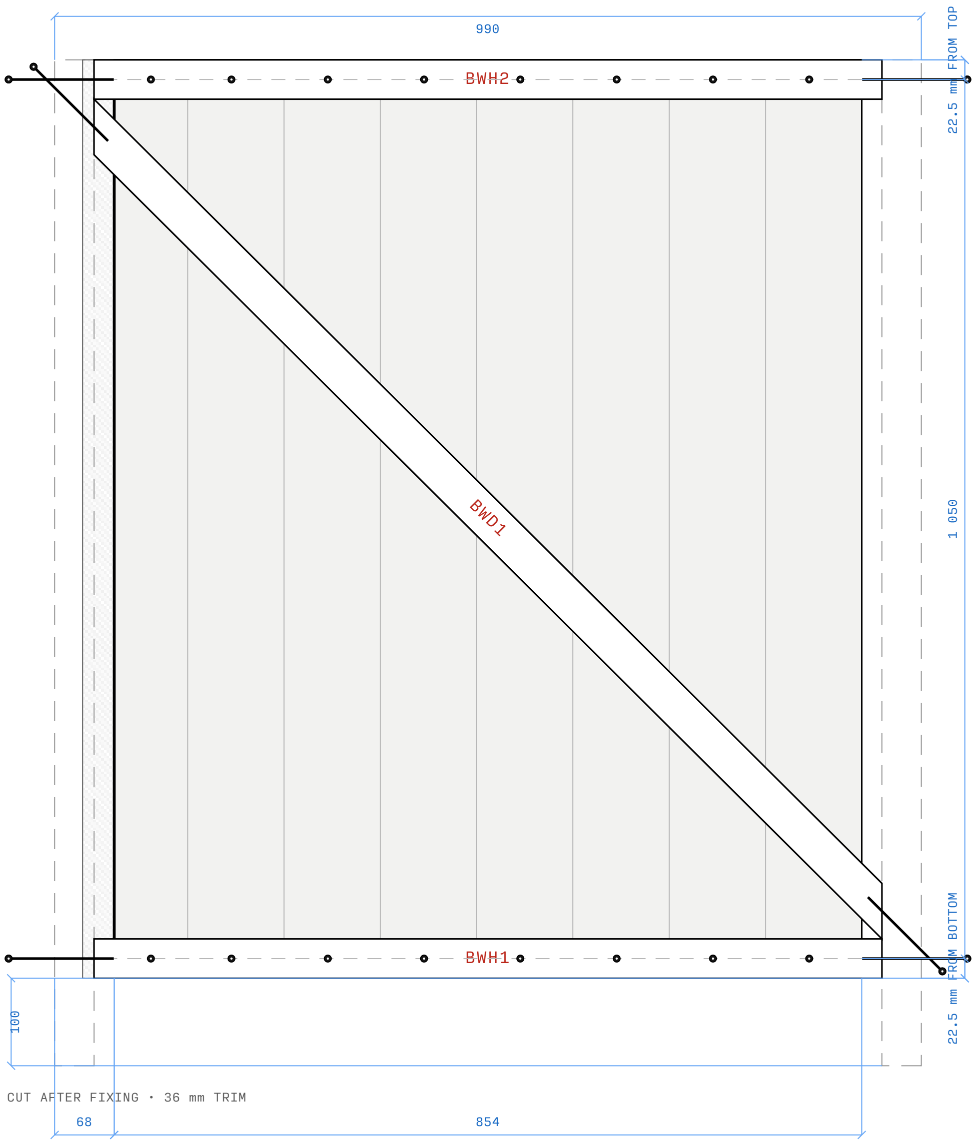
ASSEMBLY

1. Build the frame. Match its diagonals before you fix RSC1 to RSC7.
2. Fix the square RSC boards from front to rear. Align their lower and front edges.
3. Mark 1 175 at the front and 1 075 at the rear. Set the circular-saw guide and make one gang cut after the boards are fixed.
4. Use the rear frame edge to set the circular-saw guide. Cut the estimated 10 overhang, then cut the 45 × 45 bottom-front notch.
5. Do not pre-cut the roof reliefs. Hang and close the roof, then cut only to the scribe.

RSC1 RSC2 RSC3 RSC4 RSC5 RSC6 RSC7
RSD1 RSH1 RSH2 RSV1 RSV2

☐ UNIT COMPLETE

Stack E BACK UNIT



BACK UNIT · A-405 · REAR ELEVATION

seen from outside the back wall · posts ghosted · trim after the boards are fixed

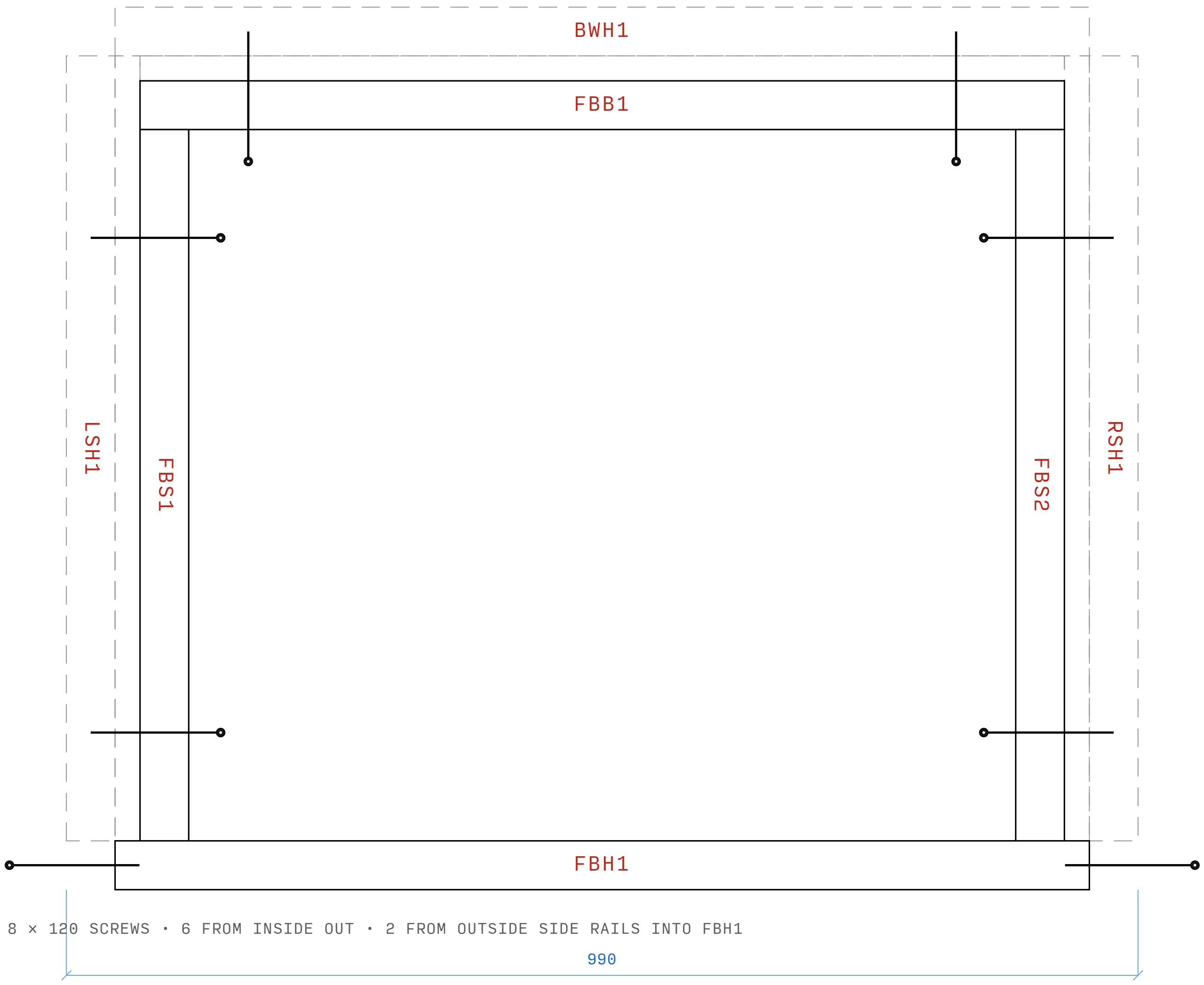
ASSEMBLY

1. Build BWH1 and BWH2 with BWD1. Install this bare frame between the two side units.
2. Attach the left and right side units to the back unit with the indicated beam screws. Drive from the rear posts into BWH1 and BWH2 at each marked landing.
3. Fix BWC1 to BWC8 inside the side cladding. Align BWC1 with the left landing mark.
4. Use the right landing mark to set the circular-saw guide. Cut the estimated 36 overhang after the boards are fixed.

BWC1 BWC2 BWC3 BWC4 BWC5 BWC6 BWC7 BWC8 BWD1 BWH1 BWH2

☐ UNIT COMPLETE

Stack F SHELL JOINT



SHELL JOINT • A-406 • PLAN

front edge at the bottom • sides and back ghosted • fit FBB1, FBS1, FBS2, and FBH1

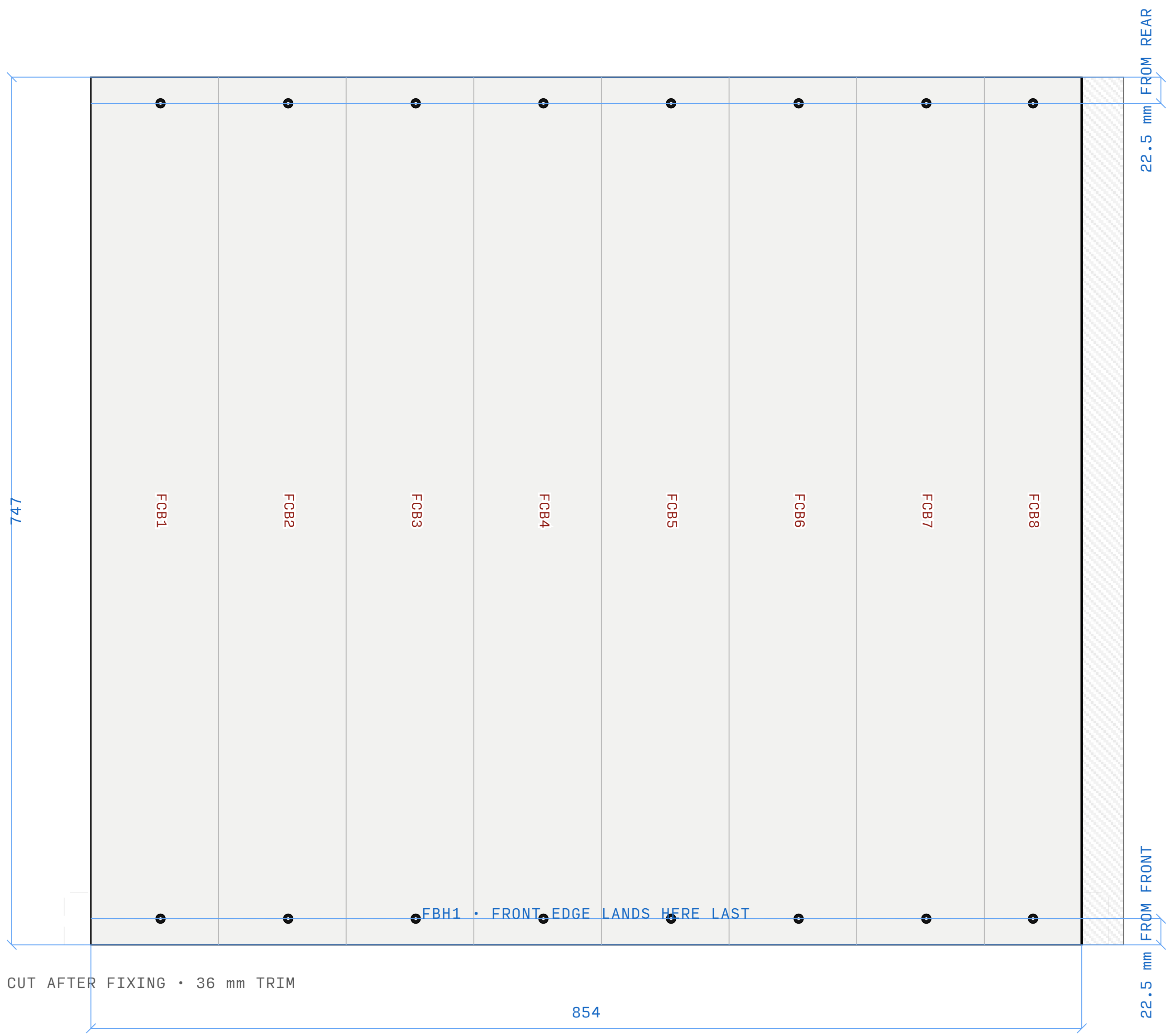
ASSEMBLY

1. Brace the left and right units upright. Keep the bottom of LSH1, RSH1, and BWH1 100 mm above ground while you install the back frame between them.
2. Fit FBB1 between BWH1, FBS1 and FBS2 between LSH1 and RSH1, and FBH1 across the front opening. Keep all four beams square and flush with the shell frame.
3. Drive two 6 × 120 mm screws from FBB1 through the back wall into BWH1, one 100 mm from each end. Drive two from FBS1 into LSH1 and two from FBS2 into RSH1, all from inside the structure outwards. Then drive one centred screw from each outside side rail into FBH1.
4. Install FBH1 across the front at the 100 leg datum. Measure the finished frame before the final screws.

FBB1 FBH1 FBS1 FBS2

☐ UNIT COMPLETE

Stack G FLOOR DECK



FLOOR DECK · A-407 · PLAN

front edge at the bottom · cladding side only · bearers already fitted

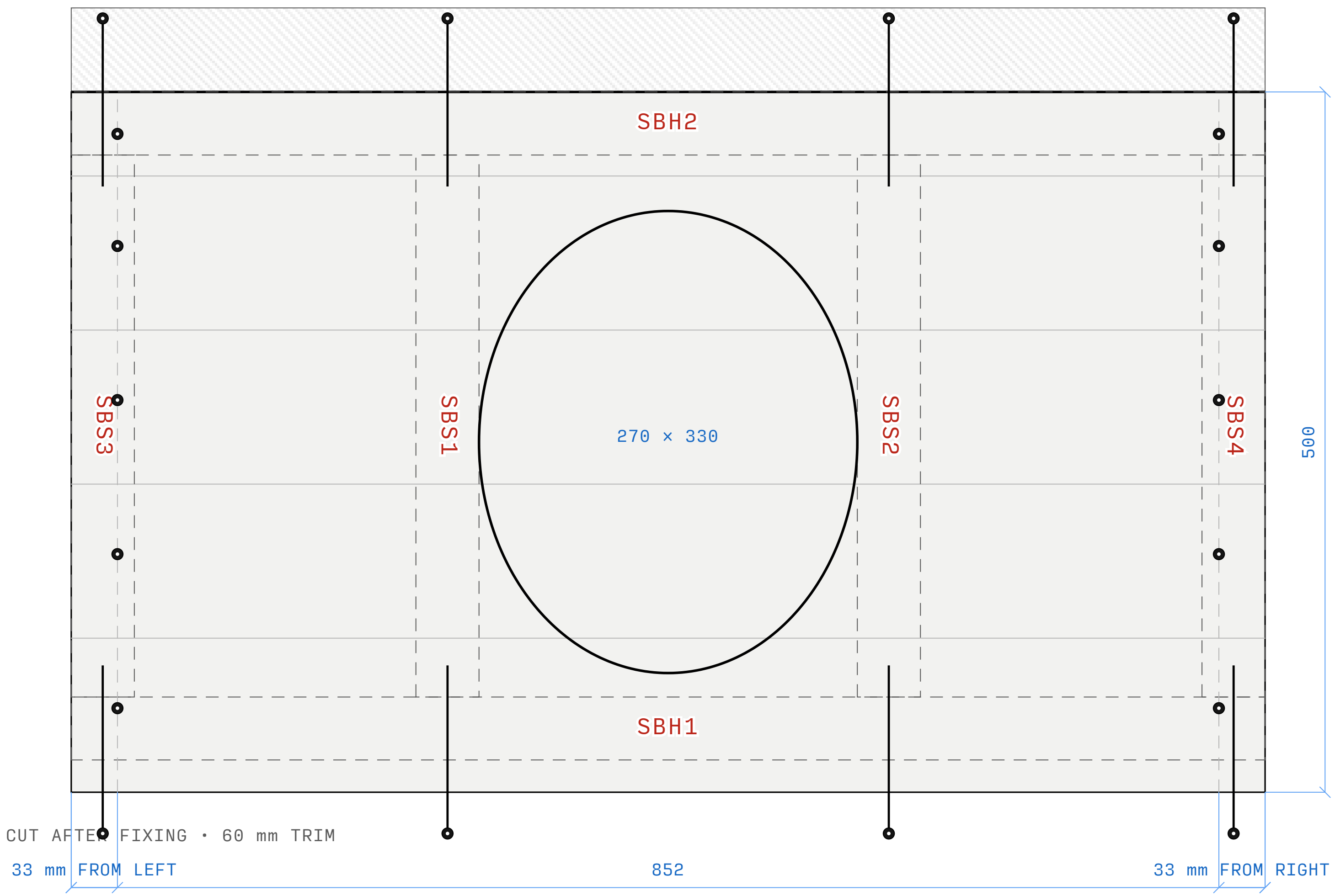
ASSEMBLY

1. Fix FCB1 to FCB8 to the installed bearers. Align FCB1 with the left bearer edge.
2. Use the right bearer edge to set the circular-saw guide. Cut the estimated 36 overhang after the boards are fixed.
3. Lower the finished deck into the square shell. Fasten its front edge to FBH1.

FCB1 FCB2 FCB3 FCB4 FCB5 FCB6 FCB7 FCB8

☐ UNIT COMPLETE

Stack H SEAT BOX



SEAT BOX · A-408 · PLAN

front edge at the bottom · 852 removable box · STB1–STB5
run across the width · trim before cutting the opening

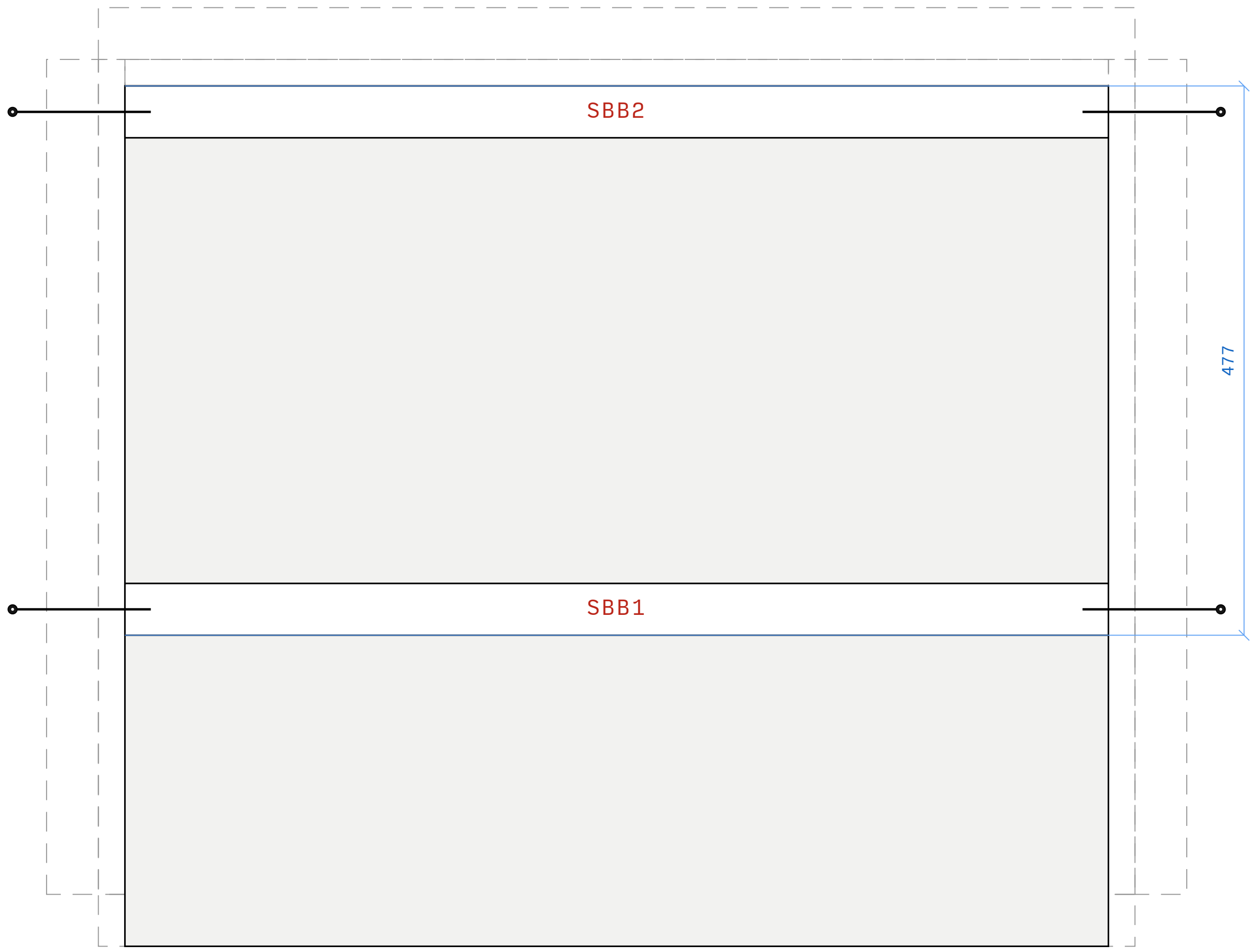
ASSEMBLY

1. Build the removable 852 mm seat box from SBH1, SBH2, SBS1–SBS4. Keep 1 mm clearance at each side.
2. Fix STB1 to STB5 across the top frame, with the board ends supported by SBS3 and SBS4.
3. Cut the estimated 60 overhang. Then cut the 270 × 330 opening.
4. Use the two outer support beams for the board screw rows. Make sure the opening clears the inner bearers and rails. Seal every fresh cladding cut.

SBH1 SBH2 SBS1 SBS2 SBS3 SBS4 STB1
STB2 STB3 STB4 STB5

☐ UNIT COMPLETE

Stack I SEAT SUPPORTS



SBB1 / SBB2 · 854 LONG · 4 × 120 SCREWS FROM OUTSIDE THROUGH SIDE CLADDING

SEAT SUPPORTS · A-409 · PLAN

front edge at the bottom · floor, sides, and back already attached · 477 from back wall to front support face

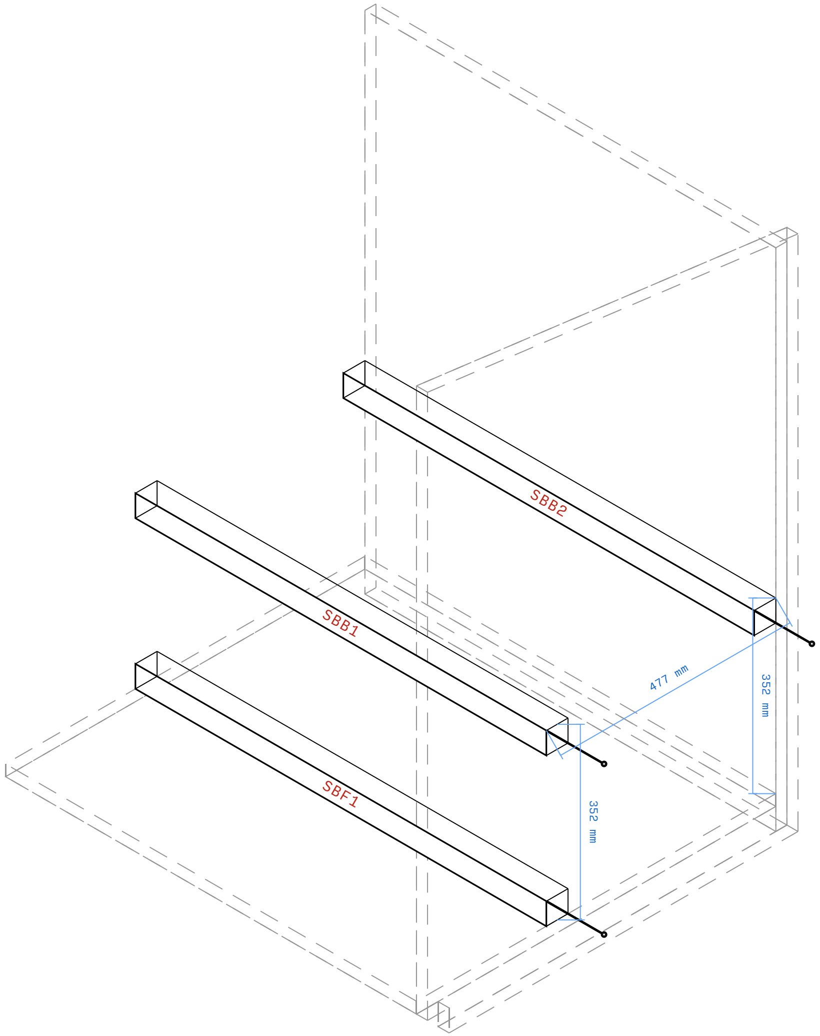
ASSEMBLY

1. With the floor deck, sides, and back attached, place SBB1 directly under SBH1 and SBB2 directly under SBH2.
2. Set the upper edge of both support beams 352 mm above the floor deck: 397 mm front-panel height minus 45 mm seat-box beam height.
3. Drive one 6 × 120 mm screw from each side exterior into the centre of each support beam. Check the front beam face against the dimension on A-409.

SBB1 SBB2

☐ UNIT COMPLETE

Stack J SEAT BOX SUPPORTS



RIGHT_WALL · SBB1 / SBB2 / SBF1 · 477 FROM BACK · 23 MM SIDE CLADDING · 2 × 120 SCREWS FROM OUTSIDE

SEAT BOX SUPPORTS · A-411 · ISOMETRIC PROJECTION

Flip the side toggle to see the two outside screw pairs.

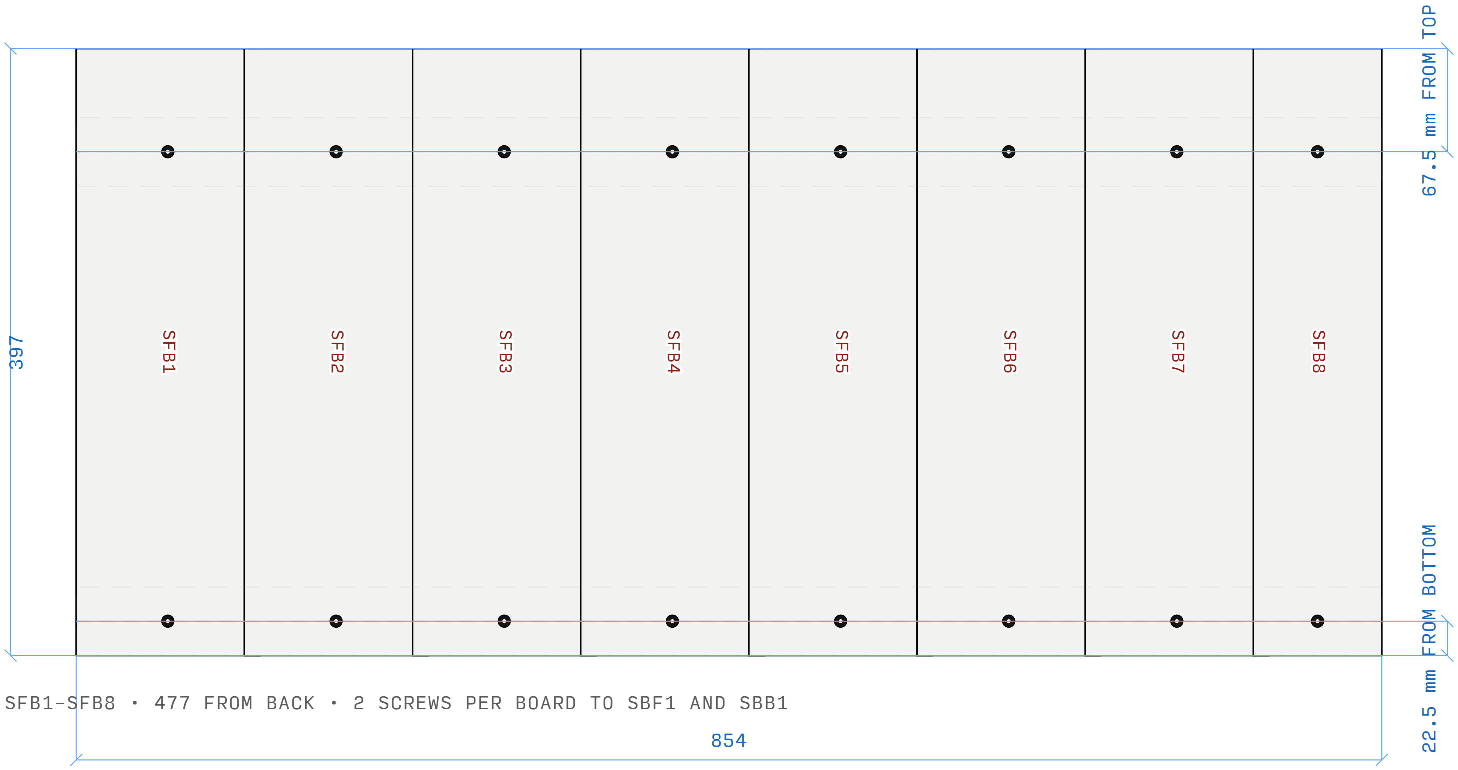
ASSEMBLY

1. Lower the centred 852 mm seat box onto SBB1 and SBB2. Leave the support beams fixed to the shell so the box can lift out as one assembly.
2. Check SBB1 and SBF1 on the shared 477 mm-from-back datum before the front boards are fixed.
3. Use the right/left toggle to check both outside screw pairs and confirm the two side clearances before final finishing.

SBB1 SBB2 SBF1

☐ UNIT COMPLETE

Stack K SEAT FRONT CLADDING



SEAT FRONT CLADDING • A-412 • FRONT ELEVATION

seen from outside the door • cladding face only • 23 mm wall • SBF1 and SBB1 ghosted

ASSEMBLY

1. Lay SFB1 to SFB8 as individual boards. Keep their vertical joints in the order shown.
2. Trim the joined field to the 854 mm before fixing.
3. Fix every board to SBF1 at floor level and SBB1 under the seat box. The two screw rows are shown on the drawing.

SFB1 SFB2 SFB3 SFB4 SFB5 SFB6 SFB7 SFB8

☐ UNIT COMPLETE